FOUR MUTUALLY UNBIASED BASES IN DIMENSION EIGHT GIVE A NON-DECOMPOSABLE WITNESS

Abstract. For $d = 2^r$ with $r > 1$, Bae, Bera, Chruściński, Hiesmayr and McNulty conjectured that the minimal number of mutually unbiased bases needed to detect bound entanglement is $m = \frac{d}{2} + 1$. We refute this at $d = 8$, inside the one witness family that statement quantifies over. We exhibit a set $\mathcal{M}_4$ of four mutually unbiased bases of $\mathbb{C}^8$ — so $m = \frac{d}{2}$, one below the conjectured minimum — and a shift $s = 1$, for which the witness $W^\Gamma(\mathcal{M}_4, s)$ of their family is *non-decomposable*. The certificate is a positive partial transpose state ρ on $\mathbb{C}^8 \otimes \mathbb{C}^8$, printed here in full as an integer matrix divided by 1049602, with

$$\operatorname{tr}\left[W^\Gamma(\mathcal{M}_4, 1)\,\rho\right] = -\frac{14588}{524801} < 0.$$

Every quantity is exact: $64\,W^\Gamma(\mathcal{M}_4, 1)$ is an integral symmetric matrix, and the positivity of ρ and of ρ^Γ is certified by exact rational $LDL^\top$ factorisations. Hence the least number of mutually unbiased bases of $\mathbb{C}^8$ for which that family has a non-decomposable member is at most 4, and not the $\frac{d}{2} + 1 = 5$ asserted at $r = 3$.

1. The conjecture

Let $\mathcal{M}_m = \{B^{(1)}, \ldots, B^{(m)}\}$ be a set of m pairwise mutually unbiased bases (MUBs) of $\mathbb{C}^d$. Spengler, Huber, Brierley, Adaktylos and Hiesmayr [2] attach to $\mathcal{M}_m$ and a shift $s \in \mathbb{Z}_d$ an operator $W^\Gamma(\mathcal{M}_m, s)$ on $\mathbb{C}^d \otimes \mathbb{C}^d$ (recalled in §2) which is nonnegative on every separable state. When $W^\Gamma(\mathcal{M}_m, s)$ is *non-decomposable*, i.e. cannot be written as $P + Q^\Gamma$ with $P, Q \succeq 0$, it detects a state with positive partial transpose, that is a bound entangled state. Bae, Bera, Chruściński, Hiesmayr and McNulty [1] asked how many MUBs are needed for this, and conjectured a sharp answer in the power-of-two dimensions.

Their Conjecture 1 reads: "*If $d = 2^r$ and $r > 1$, the minimal number of MUBs required for bound entanglement detection equals $m = \frac{d}{2} + 1 = 2^{r-1} + 1$.*"

Two clauses of [1] fix what a counterexample must be. The sentence immediately above the conjecture reads "*When $d = 2^r$, we predict that $\frac{d}{2} + 1$ MUBs are both necessary and sufficient for the witness $\mathbf{W}^\Gamma(\mathcal{M}_m, s)$ to be non-decomposable*", so the minimum is taken over the paper's own family and "detects bound entanglement" means "is non-decomposable". Second,

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the minimum is *existential*: at $d = 4$ that paper exhibits particular triples of MUBs and still calls $\frac{d}{2}+1$ minimal. One $m = \frac{d}{2}$ example at $d = 8$ is therefore the right shape of counterexample, and everything below stays inside that family.

What is settled below is the cell $r = 3$, by one exhibited object: at $d = 8$ the least m for which the family (3) has a non-decomposable member is at most $4 = \frac{d}{2}$, so it is not $\frac{d}{2} + 1 = 5$. Only the necessity half of the quoted statement is at issue; the sufficiency theorem of [1] concerns $m > \frac{d}{2} + 1$.

2. The witness family

Fix $d = 8$ and index the computational basis of $\mathbb{C}^8$ by $x = 4x_1 + 2x_2 + x_3$ with $x_1, x_2, x_3 \in \mathbb{F}_2$. For a symmetric matrix $S \in \mathbb{F}_2^{3\times 3}$ and $v \in \mathbb{F}_2^3$ put

$$|v_S\rangle = \frac{1}{\sqrt{8}} \sum_{x\in\mathbb{F}_2^3} \mathrm{i}^{\,S_{11}x_1+S_{22}x_2+S_{33}x_3}(-1)^{S_{12}x_1x_2+S_{13}x_1x_3+S_{23}x_2x_3+v\cdot x}\,|x\rangle, \tag{1}$$

where the exponent of i is read modulo 4 and the exponent of -1 modulo 2, and set $B_S = \{\,|v_S\rangle : v \in \mathbb{F}_2^3\,\}$. Each B_S is an orthonormal basis of $\mathbb{C}^8$. We encode S by the integer

$$\mathrm{idx}(S) = S_{11} + 2S_{12} + 4S_{13} + 8S_{22} + 16S_{23} + 32S_{33} \in \{0, \dots, 63\}. \tag{2}$$

Let $\mathcal{M}_m = \{B^{(1)}, \dots, B^{(m)}\}$ with $B^{(1)}$ the computational basis, and let $s \in \mathbb{Z}_8$. Writing $\overline{\,\cdot\,}$ for entrywise complex conjugation in the computational basis, the witness of [2] in the form quantified over by the conjecture is

$$\begin{aligned} W^\Gamma(\mathcal{M}_m, s) = {}& \frac{d+m-1}{d}\, I\otimes I - \sum_{l=0}^{d-1} |l\rangle\langle l| \otimes |l\oplus s\rangle\langle l\oplus s| \\ & - \sum_{k=2}^{m} \sum_{v} |v^{(k)}\rangle\langle v^{(k)}| \otimes \overline{|v^{(k)}\rangle\langle v^{(k)}|}, \end{aligned} \tag{3}$$

the inner sum running over the d elements $|v^{(k)}\rangle$ of $B^{(k)}$ and $\oplus$ denoting addition modulo d. It is nonnegative on every separable state [2].

For comparison, [1] displays the same operator as

$$\mathbf{W}^\Gamma(\mathcal{M}_m, s) = \frac{d+m-1}{d}\,\mathbf{1}_d \otimes \mathbf{1}_d - \mathbf{B}^\Gamma(\mathcal{M}_m, s),$$

$$\mathbf{B}^\Gamma(\mathcal{M}_m, s) = \sum_{\ell=0}^{d-1} |\ell\rangle\langle \ell| \otimes |\ell + s\rangle\langle \ell + s| + \sum_{\alpha=1}^{m-1}\sum_{i=0}^{d-1} |i_\alpha\rangle\langle i_\alpha| \otimes |i_\alpha^*\rangle\langle i_\alpha^*|,$$

where $\mathbf{1}_d$ is the $d\times d$ identity, $\mathcal{B}_0$ is the canonical basis, conjugation and transposition are taken with respect to $\mathcal{B}_0$, and $\ell + s$ is read modulo d. Term by term this is (3): the prefactor $\frac{d+m-1}{d}$, the shifted diagonal term and the conjugation in the second tensor factor are theirs unchanged; our $k = 2, \dots, m$ indexes their $\alpha = 1, \dots, m-1$ (the $m-1$ bases other than the computational one), our $\overline{\,\cdot\,}$ is their *, and our $\oplus$ is their $+$ modulo d.

3. The objects

$$\mathcal{M}_4 = \{B^{(1)},\, B_{S_0},\, B_{S_{13}},\, B_{S_{19}}\}, \qquad s = 1, \qquad m = 4 = \tfrac{d}{2}, \tag{4}$$

with $B^{(1)}$ the computational basis and, by (2),

$$S_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad S_{13} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad S_{19} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Lemma 1. *The four bases in* (4) *are orthonormal and pairwise mutually unbiased: all* $\binom{4}{2} \cdot 64 = 384$ *cross overlaps satisfy* $|\langle u \mid u' \rangle|^2 = \frac{1}{8}$.

Since $\sqrt{8}\,|v_S\rangle$ has entries in $\{\pm 1, \pm \mathrm{i}\}$, Lemma 1 is an identity in $\mathbb{Z}[\mathrm{i}]$: the 384 overlaps of the rescaled vectors have squared modulus exactly 8, and the 64 overlaps within a rescaled B_S are $8\delta_{vv'}$. No floating point occurs.

Lemma 2. $64\, W^\Gamma(\mathcal{M}_4, 1)$ *is a symmetric matrix of integers of absolute value at most* 64*; its imaginary part vanishes identically. Its diagonal has* 56 *entries equal to* 64 *and* 8 *equal to* 0*, so*

$$\operatorname{tr} W^\Gamma(\mathcal{M}_4, 1) = \tfrac{3584}{64} = 56 = d^2 - d.$$

Finally the state. Let $N = 1049602 = 2 \cdot 524801$ and let $M \in \mathbb{Z}^{64 \times 64}$ be the symmetric matrix whose entries on and above the diagonal are those listed in Table 1 and zero elsewhere, with $M_{ji} = M_{ij}$. Row i of the table is printed as `i | j:`M_{ij} `j:`M_{ij} `...` over the $j \geq i$ with $M_{ij} \neq 0$; the 288 listed entries extend to 512 nonzero entries of M. Put

$$\rho = \frac{1}{N}\, M. \tag{5}$$

The indices are $i = 8a + c$ with a the index in the first tensor factor and c that in the second, so the partial transpose is $(\rho^\Gamma)_{8a+c,\,8b+e} = \rho_{8a+e,\,8b+c}$.

Lemma 3. ρ *defined by* (5) *is a state with positive partial transpose:* $\rho = \rho^\top$, $\operatorname{tr} \rho = 1$ *(the* 64 *diagonal entries of* M *sum to* N*),* $\rho \succeq 0$ *and* $\rho^\Gamma \succeq 0$. *Moreover*

$$\operatorname{tr}\big[W^\Gamma(\mathcal{M}_4, 1)\, \rho\big] = -\frac{14588}{524801} = -0.0277972031303\ldots < 0.$$

Proof. Symmetry and the trace are immediate from Table 1. For the two positivity statements, the exact rational $LDL^\top$ factorisation without pivoting runs to completion on M and on M^Γ; all 64 pivots are strictly positive in each case, the smallest being

$$\frac{63834948025622562}{457387529428466097047} = 1.395643\ldots \times 10^{-4} \quad \text{for } \rho,$$
$$\frac{329439466941518560}{2324566123790645652773} = 1.417208\ldots \times 10^{-4} \quad \text{for } \rho^\Gamma,$$

TABLE 1. The 288 integers on and above the diagonal of $M = N\rho$, $N = 1049602$.

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 0 | 0:13090 9:13229 18:13171 27:12881 36:12852 45:13179 54:13190 63:12939
 1 | 1:57650 8:13270 19:5884 26:4196 37:5429 44:4029 55:-6688 62:-5143
 2 | 2:12536 11:3191 16:3951 25:-5082 38:3182 47:12324 52:-5067 61:3955
 3 | 3:12118 10:4380 17:-6578 24:3461 39:-7078 46:5584 53:10793 60:4002
 4 | 4:12962 13:3941 22:4016 31:-4767 32:-4764 41:4021 50:3890 59:12707
 5 | 5:10924 12:4029 23:10749 30:4031 33:5426 40:-5093 51:5440 58:-5084
 6 | 6:13551 15:-6379 20:-5067 29:13165 34:3425 43:3589 48:4226 57:3866
 7 | 7:10281 14:-4675 21:5450 28:4002 35:10153 42:-4621 49:5587 56:4754
 8 | 8:3911 19:4196 26:1129 37:4029 44:1076 55:-5143 62:-1587
 9 | 9:13713 18:13534 27:13171 36:13179 45:13456 54:13398 63:13190
10 | 10:26484 17:4481 24:-1699 39:5584 46:-4404 53:4106 60:16959
11 | 11:12536 16:-5082 25:3951 38:12324 47:3182 52:3955 61:-5067
12 | 12:17079 23:4031 30:16880 33:-5093 40:1209 51:-5084 58:1239
13 | 13:13566 22:-5521 31:4016 32:4021 41:-5464 50:13385 59:3890
14 | 14:16021 21:4106 28:1086 35:-4621 42:15709 49:3658 56:891
15 | 15:13551 20:13165 29:-5067 34:3589 43:3425 48:3866 57:4226
16 | 16:14089 25:4956 38:-5067 47:3955 52:4930 61:13904
17 | 17:9972 24:3967 39:10793 46:4106 53:-6110 60:2457
18 | 18:13713 27:13229 36:13190 45:13398 54:13456 63:13179
19 | 19:57650 26:13270 37:-6688 44:-5143 55:5429 62:4029
20 | 20:13245 29:-3745 34:4226 43:3866 48:4411 57:4916
21 | 21:11631 28:-5748 35:5587 42:3658 49:9750 56:-5601
22 | 22:13566 31:3941 32:3890 41:13385 50:-5464 59:4021
23 | 23:10924 30:4029 33:5440 40:-5084 51:5426 58:-5093
24 | 24:15267 39:4002 46:16959 53:2457 60:-3588
25 | 25:14089 38:3955 47:-5067 52:13904 61:4930
26 | 26:3911 37:-5143 44:-1587 55:4029 62:1076
27 | 27:13090 36:12939 45:13190 54:13179 63:12852
28 | 28:30395 35:4754 42:891 49:-5601 56:15140
29 | 29:13245 34:3866 43:4226 48:4916 57:4411
30 | 30:17079 33:-5084 40:1239 51:-5093 58:1209
31 | 31:12962 32:12707 41:3890 50:4021 59:-4764
32 | 32:12962 41:3941 50:4016 59:-4767
33 | 33:10924 40:4029 51:10749 58:4031
34 | 34:13551 43:-6379 48:-5067 57:13165
35 | 35:10281 42:-4675 49:5450 56:4002
36 | 36:13090 45:13229 54:13171 63:12881
37 | 37:57650 44:13270 55:5884 62:4196
38 | 38:12536 47:3191 52:3951 61:-5082
39 | 39:12118 46:4380 53:-6578 60:3461
40 | 40:17079 51:4031 58:16880
41 | 41:13566 50:-5521 59:4016
42 | 42:16021 49:4106 56:1086
43 | 43:13551 48:13165 57:-5067
44 | 44:3911 55:4196 62:1129
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47 | 47:12536 52:-5082 61:3951
48 | 48:13245 57:-3745
49 | 49:11631 56:-5748
50 | 50:13566 59:3941
51 | 51:10924 58:4029
52 | 52:14089 61:4956
53 | 53:9972 60:3967
54 | 54:13713 63:13229
55 | 55:57650 62:13270
56 | 56:30395
57 | 57:13245
58 | 58:17079
59 | 59:12962
60 | 60:15267
61 | 61:14089
62 | 62:3911
63 | 63:13090
```

after division by N. A Hermitian matrix admitting an $LDL^\top$ factorisation with all pivots positive is positive definite. The trace is the integer pairing $\sum_{i,j}(64\,W^\Gamma)_{ij}M_{ji}$ divided by $64N$; the numerator is -1867264, and $-1867264/(64 \cdot 1049602) = -14588/524801$. The largest denominator occurring in ρ is N itself, and every entry of M has absolute value at most 57650. □

4. The result

Theorem 4. *$W^\Gamma(\mathcal{M}_4, 1)$ of (4) is non-decomposable. Consequently $m = 4 = \frac{d}{2}$ mutually unbiased bases of $\mathbb{C}^8$ detect bound entanglement through (3), the least number of mutually unbiased bases of $\mathbb{C}^8$ for which (3) has a non-decomposable member is at most 4, and the conjecture of [1] quoted in §1, which asserts $\frac{d}{2}+1 = 5$ at $d = 8$, is false at $r = 3$.*

Proof. Suppose $W^\Gamma(\mathcal{M}_4, 1) = P + Q^\Gamma$ with $P, Q \succeq 0$. For any state σ with $\sigma^\Gamma \succeq 0$,

$$\mathrm{tr}\big[W^\Gamma(\mathcal{M}_4,1)\sigma\big] = \mathrm{tr}[P\sigma] + \mathrm{tr}\big[Q^\Gamma\sigma\big] = \mathrm{tr}[P\sigma] + \mathrm{tr}\big[Q\,\sigma^\Gamma\big] \;\geq 0,$$

because the trace of a product of two positive semidefinite operators is nonnegative and $\mathrm{tr}[XY^\Gamma] = \mathrm{tr}[X^\Gamma Y]$. By Lemma 3 the state ρ has $\rho^\Gamma \succeq 0$ and $\mathrm{tr}[W^\Gamma(\mathcal{M}_4,1)\rho] < 0$, so no such P, Q exist.

By [2] the operator $W^\Gamma(\mathcal{M}_4, 1)$ is nonnegative on separable states, so $\mathrm{tr}[W^\Gamma(\mathcal{M}_4,1)\rho] < 0$ forces ρ to be entangled; since $\rho^\Gamma \succeq 0$, it is bound entangled. It is detected by the witness built from the four MUBs of (4), and Lemma 1 confirms that these are four pairwise mutually unbiased bases. Hence four MUBs suffice at $d = 8$, and the minimum over the family (3) is at most $4 < 5 = \frac{d}{2} + 1$. □

5. Scope

Only the cell $r = 3$ is treated; dimensions $d = 2^r$ with $r \geq 4$ are untouched, and no pattern is proposed in place of the refuted one. What is proved is "at most 4", not "exactly 4": whether $m = 3$ or $m = 2$ also suffice at $d = 8$ is not addressed. Only the particular $\mathcal{M}_4$ and the shift $s = 1$ of (4) are certified, and nothing here asserts that every set of four mutually unbiased bases of $\mathbb{C}^8$, or every shift, gives a non-decomposable witness. The number $-\frac{14588}{524801}$ is the value at the ρ printed above, not the minimum of $\mathrm{tr}[W^\Gamma(\mathcal{M}_4,1)\sigma]$ over PPT states σ; only its sign is used in Theorem 4. That $W^\Gamma(\mathcal{M}_m, s)$ is nonnegative on separable states is the defining property established in [2]; it is used, not reproved, here.

6. Reproduction and verification

Everything needed is printed above: the index conventions (1) and (2), the three symmetric matrices S_0, S_{13}, S_{19}, the shift $s = 1$, the witness (3), and the 288 integers of Table 1. From these, $64\,W^\Gamma(\mathcal{M}_4, 1)$ is assembled in

$\mathbb{Z}[\mathrm{i}]$ — the rescaled basis vectors $\sqrt{8}\,|v_S\rangle$ have entries in $\{\pm 1, \pm \mathrm{i}\}$, and each summand of (3) contributes $u_a \bar{u}_b \bar{u}_c u_e$ to the entry indexed by $\big((a,c),(b,e)\big)$ — and the four facts of Lemma 3 are three integer computations and two rational $LDL^\top$ factorisations.

The calculation was re-run independently, in exact integer and rational arithmetic throughout, from the objects as printed above and from nothing else. The program rebuilds the four bases from (1), checks orthonormality and the 384 cross overlaps of Lemma 1, assembles $64\,W^\Gamma(\mathcal{M}_4, 1)$ and verifies Lemma 2, parses Table 1, and verifies Lemma 3: the trace, both $LDL^\top$ factorisations with their smallest pivots, and $\mathrm{tr}[W^\Gamma(\mathcal{M}_4, 1)\rho]$ by two independent routes. All of its checks passed. The program and its transcript accompany this note; they are not needed in order to redo the calculation, since every input they consume is printed above.

References

[1] J. Bae, A. Bera, D. Chruściński, B. C. Hiesmayr, and D. McNulty, *How many mutually unbiased bases are needed to detect bound entangled states?*, J. Phys. A **55** (2022), no. 50, 505303. doi:10.1088/1751-8121/acaa16; arXiv:2108.01109.

[2] C. Spengler, M. Huber, S. Brierley, T. Adaktylos, and B. C. Hiesmayr, *Entanglement detection via mutually unbiased bases*, Phys. Rev. A **86** (2012), 022311. doi:10.1103/PhysRevA.86.022311.